# Practical

## SUMMARY:-

This practical demonstrated the implementation of Linear Search and Binary Search in C. It helped understand how both algorithms search for an element and compare their efficiency.

**Linear Search** checks each element one by one until the target element is found or the array ends.

### Output

```
number is found

...Program finished with exit code 0
Press ENTER to exit console.
```

### Time Complexity

| Case | Time Complexity |
|---|---|
| Best Case | O(1) |
| Average Case | O(n) |
| Worst Case | O(n) |

**Binary Search** repeatedly divides a sorted array into two halves to quickly find the target element.

### Output

```
the number is found:40

...Program finished with exit code 0
Press ENTER to exit console.
```

## Time Complexity

| Case | Time Complexity |
| --- | --- |
| Best Case | O(1) |
| Average Case | O(log n) |
| Worst Case | O(log n) |

## Conclusion :

We learned that Linear Search works on any array but is slower, while Binary Search is faster and more efficient for sorted arrays.